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Shock tube/laser absorption measurements of ethylene time-histories during ethylene and *n*-heptane pyrolysis

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Abstract

Ethylene concentration time-histories were measured behind reflected shock waves in ethylene/argon and *n*-heptane/argon mixtures using 10.5 μm CO_2 laser absorption. Reflected shock conditions covered temperatures between 1350 and 1950 K, pressures between 1.3 and 3.3 atm, and fuel concentrations of 1% ethylene and 300 ppm *n*-heptane in argon. The ethylene absorption cross-section at this wavelength was determined over similar pressures and temperatures using both absorption measurements in a heated-cell FTIR and laser absorption/shock tube measurements. Measured ethylene concentration time-histories during ethylene pyrolysis were compared to constant volume simulations using the Marinov et al. (1998) mechanism. A sensitivity of the measured profiles to the rate constant of the primary ethylene decomposition reaction, $C_2H_4 \rightarrow C_2H_2 + H_2$ was demonstrated, allowing a determination of the rate constant for this reaction using the ethylene time-histories. Ethylene concentrations, measured during *n*-heptane pyrolysis, were also compared to simulations using the Sirjean et al. (2009)/JetSurF 1.0 mechanism. Incorporation of the new reaction rate constant k_1 provided better agreement between the measured ethylene concentration profiles and the simulations for both ethylene and *n*-heptane pyrolysis.
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Keywords: Infrared laser absorption; Ethylene; *n*-Heptane; CO_2 laser

1. Introduction

Olefins, such as ethylene, propene and butene, are important intermediate species that form during the oxidation and pyrolysis of large alkanes [1–3]. They are also primary species involved in the formation of benzene, a key precursor of soot [4–6]. Accurate sub-mechanisms describing the formation and removal of alkenes are thus needed for high-fidelity alkane pyrolysis and oxidation mechanisms, and validation of these sub-mechanisms requires quantitative measurements of alkene species time-histories. Because of its high-temperature stability, ethylene is often the dominant alkene intermediate formed during alkane pyrolysis and oxidation, and hence, species time-history data describing this intermediate are particularly needed.

Direct measurement of ethylene time-histories can be achieved in shock tube experiments using laser absorption diagnostics. Laser diagnostics are generally superior to other methods because they are nonintrusive, can be species-specific and sensitive, and can have fast response times. Mid-IR lasers have been used to measure various hydrocarbons such as alkanes and alkenes, since these species have strong absorption bands in this range due to the C–H vibrational stretch mode near 3.4 μm [7]. However, accurate measurements of ethylene in the mid-IR during hydrocarbon

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pyrolysis or oxidation are complicated by the presence of interfering absorption from other hydrocarbon intermediates. Minimizing these interference problems requires finding another spectral region that is more specific to ethylene. In the region of the CH_2 vibrational wagging mode near 10.5 μm, typical of alkenes with a double bond at one end of the molecule, the ethylene absorption signal is about five times stronger than in the mid-IR region at room temperature [8,9]. This feature is composed of the interaction of three vibrational modes of ethylene, namely the ν_7, ν_4 and ν_{10} modes. Absorption by ethylene in this region is dominated by the ν_7 band near 10.5 μm [10]. Furthermore, in this wavelength region ethylene is the main absorber for species likely to be found during alkane oxidation and pyrolysis.

The greater part of the 10.5 μm ethylene absorption spectrum occurs in the same region as the emission from a CO_2 gas laser; and fortuitously, the Q-branch of the ν_7 ethylene band at 10.532 μm has a strong overlap with the P_{14} CO_2 line. Using a CO_2 laser at this wavelength thus offers an excellent strategy for measuring ethylene time-histories while reducing interferences from other species involved in combustion processes.

Because ethylene occurs as an atmospheric pollutant produced by industrial processes, spectroscopic data at atmospheric pressure and room temperature is generally available for this transition [11–15]. However, to develop this measurement technique for combustion processes, the cross-section database of ethylene has to be extended over a wider range of pressures and temperatures.

Here we first describe the development of an IR ethylene laser-absorption diagnostic, including determination of the absorption cross-section at elevated temperatures and pressures. We then demonstrate applicability of this diagnostic to measurements of ethylene time-histories during ethylene and *n*-heptane pyrolysis, and the use of these data to measure the rate constant of the dominant ethylene decomposition reaction, $C_2H_4 \rightarrow C_2H_2 + H_2$.

2. Experimental apparatus

Ethylene absorption cross-sections and species time-histories were measured in a shock tube at elevated temperatures, and in a room temperature static cell, using IR laser absorption at 10.532 μm. To extend this database at lower temperatures (below 700 K), absorption cross-sections of ethylene were also measured in a heated cell using an FTIR.

FTIR absorption measurements were performed with a Nicolet 6700 FTIR spectrometer equipped with an XT-KBr beam splitter, IR source, and an HgCdTe detector; the overall system has a manufacturer-specified maximum resolution of 0.09 cm^{-1} (FWHM). Linearization of the detector was accomplished through factory-calibrated electronics. The test gas cell was completely enclosed inside an oven to provide uniform heating and the temperature was monitored using a thermocouple.

High temperature (above 600 K) ethylene cross-sections and species time-histories were measured behind incident and reflected shock waves in an unheated high-purity shock tube. This shock tube has an inner diameter of 15.24 cm. The driven section is made of electro-polished stainless steel with a length/diameter (L/D) ratio of 70 and used a helium-filled driver section with an L/D of 30 to burst 0.25 mm thick polycarbonate diaphragms. The shock tube is sealed with Viton O-rings, is operated at room temperature, and can be turbo-pumped overnight to an ultimate pressure lower than 10^{-6} torr and to a leak and out-gassing rate lower than 3×10^{-6} torr/min. Typically reflected shock wave experiments were performed hourly, and over this time frame the ultimate shock tube pressure before filling was $\sim 6 \times 10^{-6}$ torr and the leak and out-gassing rate was $\sim 10^{-5}$ torr/min.

The stainless steel mixing assembly used with the shock tube facility included a multi-valve manifold and a 40 l holding tank with a magnetically driven stirring vane; both the manifold and the tank were heated uniformly to 60 °C. Mixtures were made manometrically using capacitance manometers (MKS Baratron) with a relative uncertainty of ±1%.

Incident shock speeds were determined by measuring the incident shock arrival times using a series of five piezo-electric pressure transducers (PZT) (PCB model 134) spaced at approximately 30 cm intervals over the last meter of the driven section and Phillips/Fluke PM5555 interval counters. Incident shock speeds were extrapolated to the end wall position and reflected shock conditions were calculated using an in-house-developed chemically frozen shock code (FROSH) and the thermodynamic database from Kee et al. [16] The resulting uncertainty in reflected shock temperatures over test times of 1 ms (typical for these experiments) was ±1% or 16 K at $T_5 = 1600$ K. Uncertainty in the reflected shock pressure P_5 was also ±1% and was confirmed with equal uncertainty using a sidewall PZT pressure measurement (Kistler model 601B1).

The laser used in these experiments was a grating-tuned CO_2 gas laser (Model Lasy-4G, Access Laser Co.). Carbon dioxide has three vibrational modes: the *symmetric* mode ν_1, the *bending* mode ν_2, and the *asymmetric* mode ν_3. The 10.5 μm lasing region corresponds to an emission transition from the $(\nu_1\ \nu_2\ \nu_3) = (0\ 0\ 1)$ vibrational level to the (1 0 0) level. Classical ro-vibrational transition rules apply for this transition, i.e., $\Delta J = +1$ for the

P-branch and $\Delta J = -1$ for the R-branch. The lasing transition is thus either P_i or R_i, with i being the lower rotational level. In the following experiments, the P_{14} line at 10.532 μm was used.

Preliminary experiments have shown that, under certain conditions, this CO_2 laser can emit two wavelengths simultaneously. To prevent this, a monochromator has been used to filter out any unwanted wavelength (Fig. 1). The monochromator used has a Fastie–Ebert configuration with a focal length of 50 cm. The grating has 75 groves/mm and is blazed at 7 μm, which (owing to the large amount of laser power) diffracts sufficient light at 10.5 μm. Both entrance and exit slits are 200 μm. The spatial resolution of the spectrometer with these slits is 1.0 cm^{-1}. The spacing of CO_2 lines in the 10.5 μm region is 1.72 cm^{-1} and hence is larger than the monochromator resolution [17].

Liquid-nitrogen-cooled HgCdTe detectors (IR Associates MCT-12.5, 2 MHz bandwidth, optimized for 10.5 μm) were used to detect light. The detector size is 4 mm × 4 mm. All measurements were performed at a location 2 cm from the end wall of the shock tube through barium fluoride (BaF_2) windows 1.27 cm in diameter and 2 mm thick.

Common mode rejection (CMR) was used to reduce laser intensity noise to less than 1%. CMR was performed by ratioing or subtracting the two detector signals. One detector records the signal going through the gas sample, and the other records a reference signal directly from the laser to filter intensity fluctuations. Over 5 ms the absorbance detection limit of our system is 0.5%; over 2 min it is 1%.

Ethylene concentration is derived from the absorption measurement using the Beer–Lambert law:

$$I/I_0 = \exp(-\sigma_\lambda NL) \quad (1)$$

where I_0 is the reference intensity, I is the transmitted intensity, σ_λ is the absorption cross-section at wavelength λ (cm^2/mol), N is the ethylene concentration (mol/cm^3) and L is the path length (cm). The combined quantity $\sigma_\lambda NL$ is termed the absorbance.

3. Ethylene cross-section measurements

Room temperature FTIR and laser absorption measurements were performed at 1 atm, with a 0.25% ethylene in argon mixture. Results were compared with the quantitative measurements from Pacific Northwest National Laboratories (PNNL [9]). Ethylene cross-section values at 10.532 μm were 92.25 m^2/mol for the FTIR measurement, 91.80 m^2/mol for the CO_2 laser measurement, and 90.68 m^2/mol from the PNNL data. Both FTIR and laser techniques show very good agreement with PNNL, within 2%. Differences between the current study and PNNL values can be attributed, at least partially, to the lower resolution of the PNNL study.

3.1. Ethylene absorption cross-section (300–2000 K)

Measurements of the ethylene absorption cross-section at 10.532 μm were made between 300 and 700 K using the FTIR, and between 600 and 2000 K using CO_2 laser absorption behind both the incident and reflected shock wave in the shock tube. This range of temperatures allows an overlap region of 600–700 K. Results are presented in Fig. 2.

FTIR experiments were performed using 0.25% ethylene in argon mixture at 1 atm. Shock tube experiments were performed using 0.5%, 1%, and 2% ethylene in argon mixtures. Each shock wave experiment provided two cross-section measurements: a lower temperature measurement behind the incident shock wave and a higher temperature measurement behind the reflected

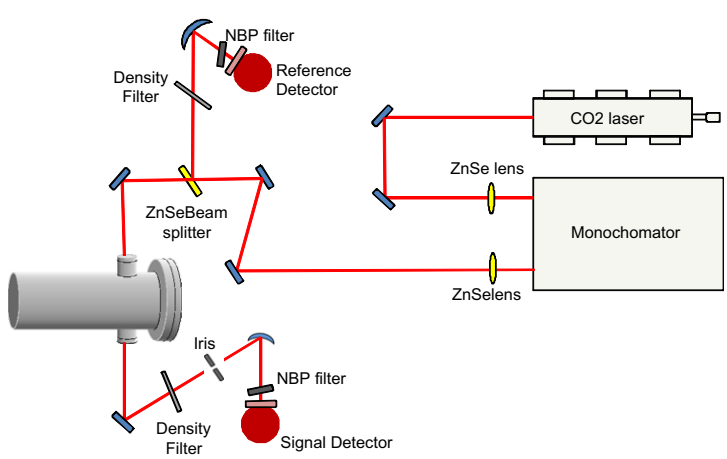

Fig. 1. Shock tube experimental setup.

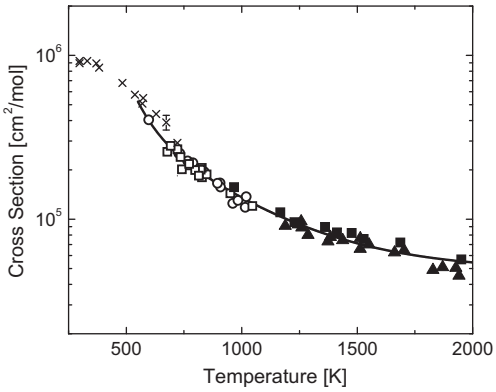

Fig. 2. Ethylene cross-sections at 10.532 μm. FTIR measurements: crosses, 1 atm; CO_2 laser absorption measurements behind incident shock waves: hollow circles, 0.35–0.7 atm; hollow squares, 0.7–1.2 atm; behind reflected shock waves: solid squares, 1.45–2.5 atm; solid triangles, 2.6–3.7 atm. Curve fit is given by Eq. (2).

shock wave. The typical pressure and temperature ranges for the measurements behind incident shock waves were 0.35–1.2 atm and 600–1100 K, and behind reflected shock waves were 1.45–3.7 atm and 950–1950 K.

In a first approximation, the ethylene cross-section can be considered a function of temperature only. A good mathematical fit of our shock tube data is a double-exponential fit:

$$\sigma_T = \sigma_0 + A\exp(-T/\alpha) + B\exp(-T/\beta)\ (\mathrm{cm^2/mol}) \tag{2}$$

where $\sigma_0 = 48520$, $A = 1.3 \times 10^6$ and $B = 3.1 \times 10^8$, in $\mathrm{cm^2/mol}$; $\alpha = 373.7$ K and $\beta = 74.2$ K. This is a purely mathematical fit to the shock tube data from 575 to 1950 K.

At higher temperatures of interest to the present study, i.e., above 1000 K, some pressure dependence is evident in the shock tube data and can be modeled by assuming that $\sigma_{T,P}$ is a product of two independent functions for T and P:

$$\sigma_{T,P} = \sigma_T \times h(P)\ (\mathrm{cm^2/mol}) \tag{3}$$

A plot of the function $h(P)$ (not shown here) shows a weak monotonic pressure dependence over the temperature range from 1000 to 1950 K. A linear least-squares fit to the data over this temperature range is given by the expression: $h(P) = 1.224–0.085\ P$ (atm). $\sigma_{T,P}$ is used to convert all ethylene absorbance traces to mole fraction in this study and was found to have an uncertainty (2σ) of ±5%.

3.2. Interference species

Measurements of interference-species absorption were performed at 10.532 μm at high temperature and pressures of 2.5–3 atm (with the experimental setup presented in Fig. 1). Species investigated include propene, 1-butene, ethane, methane, H_2O and CO_2; the resulting cross-sections are presented in Fig. 3. The largest interferences, propene and butene, have five times lower absorption cross-sections than ethylene. This smaller absorption cross-section combined with significantly lower product yields from decomposition typically leads to effectively negligible interference from these species. Methane and ethane cross-sections are about two orders of magnitude lower than ethylene, confirming that alkane species do not significantly interfere with the ethylene measurement at 10.532 μm.

Acetylene is a major product of ethylene decomposition. According to studies at PNNL [9], acetylene presents no absorption at this wavelength. Our measurements of ethylene decomposition (presented in the next section) show no absorption at high temperature at long times (after 1 ms when acetylene formation is assumed to have occurred), confirming that acetylene also does not absorb significantly at 10.532 μm at high temperature.

4. Ethylene time-histories

4.1. Ethylene pyrolysis

Analysis of the ethylene pyrolysis data was performed using the Marinov et al. (1998) mechanism [18]. In this mechanism, ethylene is removed through two main channels:

$$C_2H_4 \leftrightarrow C_2H_2 + H_2 \qquad \text{(Rxn. 1)}$$

$$C_2 + H \leftrightarrow C_2H_3 + H_2 \qquad \text{(Rxn. 2)}$$

Reaction rate constants for the decomposition of ethylene (Rxn. 1) were determined from ethylene time-history measurements in neat ethylene -pyro-

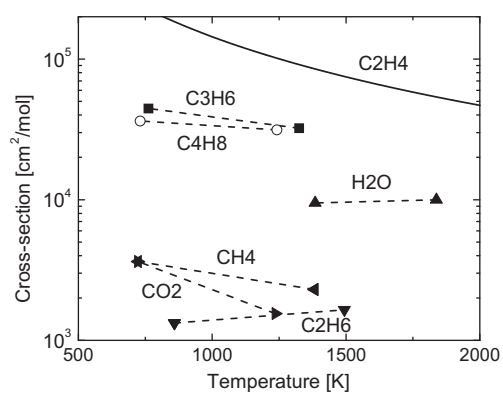

Fig. 3. Absorption cross-sections for selected interfering species at 10.532 μm. All measurements performed at 2.5–3.0 atm.

lysis experiments in the temperature range 1390–1870 K and in the pressure range 2.3–3.1 atm. Sensitivity analyses (Fig. 4) at two temperatures, 1870 and 1665 K, shows that at early times, ethylene concentrations are sensitive to the ethylene decomposition rate constant. We can take advantage of this dominant sensitivity to determine the reaction rate constant k_1 in this temperature range.

The rate constant of Rxn. 1, k_1, was determining by best-fitting ethylene time-histories over the early part of the absorption trace by varying only k_1. Representative data, best fits, and the original Marinov et al. simulations are shown in Fig. 5. The best-fit rate constant expression for k_1 over the temperature and pressure ranges of 1390–1870 K and 2.3–3.1 atm was found to be:

$$k = 5.5 \times 10^{14} \exp(-27,300/T, \mathrm{K}) \pm 35\% \quad (\mathrm{cm}^3/\mathrm{mol/s}).$$

The uncertainty in this measurement (±35%) is dominated by the effect of the temperature uncertainty (±1%).

An Arrhenius plot for k_1 is shown in Fig. 6. The current measurements fall between the extrapolation of the shock tube schlieren measurements of Kiefer et al. [21] and the UV lamp measurements of Zelson et al. [19] and ARAS measurements of Just et al. [20]. The expression for the rate constant overlaps the rate constant used by Marinov et al. (calculated using the Lindemann form with no specific collider efficiency for argon) [18]. The availability of ethylene time-histories now opens up the possibility of further investigations and improvements to the ethylene pyrolysis sub-mechanism. In particular, a study of the high-pressure decomposition of ethylene is possible.

4.2. n-Heptane pyrolysis

n-Heptane has been extensively used as a surrogate fuel to study diesel combustion chemistry, first because its cetane index of 56 is similar to that of diesel, and in addition, because there are relatively well-established kinetic reaction mechanisms available for this fuel. The ethylene measurement technique presented in the first part of this paper has been used to measure ethylene time-histories during *n*-heptane pyrolysis.

Measurements were performed behind reflected shock waves using the experimental setup described previously. Experiments covered a temperature range of 1247–1874 K and a pressure range of 2.6–3.2 atm. The test gas mixture was 300 ppm *n*-heptane in argon and were confirmed (to within 5%) using *in situ* 3.39 μm laser absorption [22]. Figure 7 presents ethylene time-histories for four *n*-heptane pyrolysis experiments at 1247, 1370, 1662, and 1874 K.

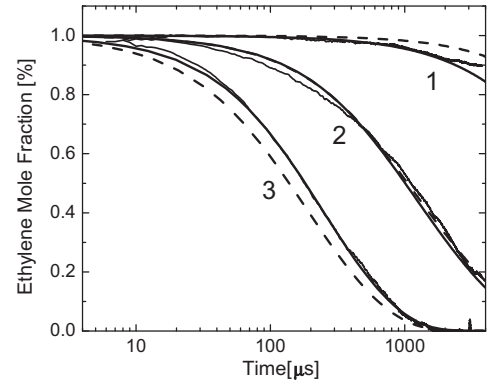

Fig. 5. Representative ethylene time-histories during ethylene pyrolysis behind reflected shock waves. Example 1: 1392 K, 3.14 atm; example 2: 1665 K, 2.78 atm; example 3: 1870 K, 2.66 atm. The mixture in all three examples was 1% ethylene in argon. Dashed lines: simulations performed using constant *U–V* modeling with the Marinov et al. mechanism [18]; heavy solid lines: Marinov et al. mechanism with best-fit k_1; light solid line: experimental data.

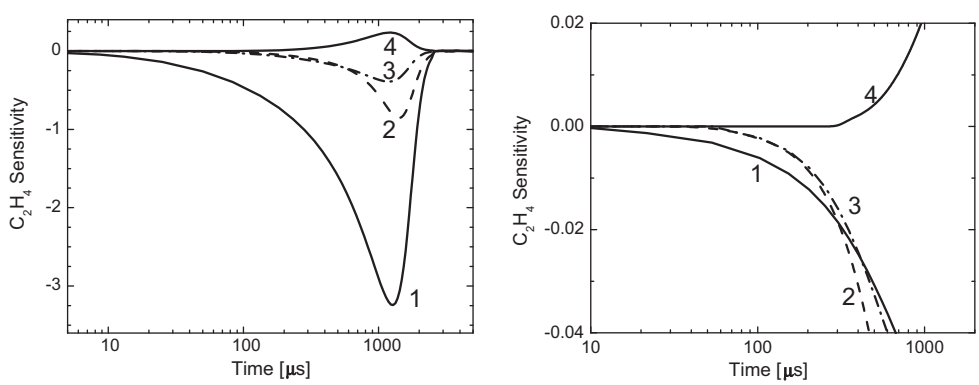

Fig. 4. Ethylene sensitivity analysis for 1% C_2H_4 in Ar. Initial reflected shock conditions: left panel: 1870 K, 2.66 atm; right panel: 1665 K, 2.78 atm. Labels for the four reactions with largest sensitivities: (1) Rxn. 1; (2) Rxn. 2; (3) $C_2H_4 + M = C_2H_3 + H + M$; (4) $C_2H_3 + H = C_2H_2 + H_2$.

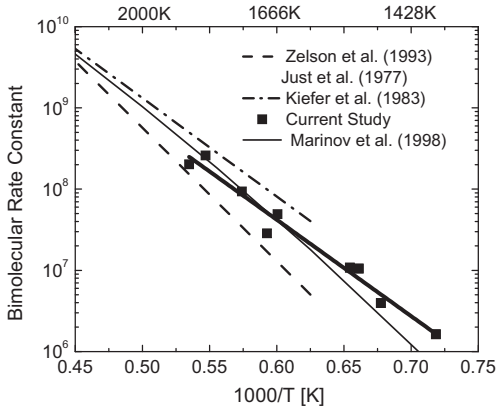

Fig. 6. Arrhenius plot for Reaction 1: $C_2H_4 \rightarrow C_2H_2 + H_2$. Solid squares/heavy line: current study; dashed line: Zelson et al. [19] and Just et al. [20]; thin solid line: Marinov et al. [18] model (3 atm); dot–dashed line: extrapolation of Kiefer et al. [21].

Simulations performed with the JetSurF 1.0 mechanism from Sirjean et al. (2009) [23] were compared to experimental results, as shown in Fig. 7. In the two higher temperature examples of Fig. 7, after its rapid formation to a plateau levels, ethylene starts to decompose during the 2 ms measurement test time, and this decomposition is controlled by the same processes as in neat ethylene pyrolysis. JetSurF 1.0 uses an ethylene submechanism that is similar to the Marinov et al. mechanism, however, in JetSurF 1.0, Rxn. 1 is divided into two steps:

$$C_2H_4 + M \leftrightarrow H_2CC + H_2 + M \qquad \text{(Rxn. A)}$$

$$H_2CC + M \leftrightarrow C_2H_2 + M \qquad \text{(Rxn. B)}$$

Rxns. A and B yield effectively the same products as Rxn. 1. We replaced the rate constant of Rxn. A in the JetSurF 1.0 mechanism with the updated value of k_1 determined in our ethylene pyrolysis study and left the rate constant for Rxn. B unchanged. The resulting mechanism is called updated JetSurF 1.0. This updated mechanism provides an improved prediction of ethylene removal at the two higher temperatures shown: 1662 and 1874 K. Thus, the ethylene decomposition rate constants found in the ethylene pyrolysis experiments (described in the previous section) are also consistent with the ethylene time-histories of these *n*-heptane pyrolysis experiments.

The plateau yields of ethylene during *n*-heptane pyrolysis are in agreement with JetSurF 1.0 predictions for all temperatures. However, measured rates of ethylene formation during *n*-heptane pyrolysis are slightly faster than the JetSurF 1.0 simulation rates, as seen in the two lower temperature examples in Fig. 7. A sensitivity analysis shown in Fig. 8 indicates, as expected, that ethylene concentrations are strongly sensitive to the rate constants of both *n*-heptane decomposition and to H-atom abstraction from *n*-heptane. Increases in one or more of these rate constants would be needed to improve ethylene time-history simulations at early times in these lower temperature examples.

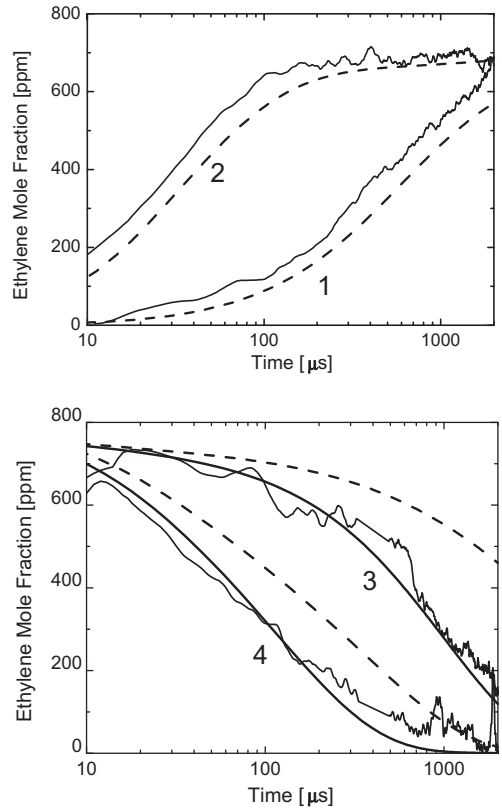

Fig. 7. Ethylene time-histories during *n*-heptane pyrolysis. Example 1: 1247 K, 3.2 atm; example 2: 1370 K, 3.1 atm; example 3: 1662 K, 2.83 atm; example 4: 1874 K, 2.67 atm. Test gas mixture for all examples is 300 ppm *n*-heptane in argon. Small oscillations in the measurements at high temperatures are a result of low signal-to-noise ratio due to the small absorption cross-section for ethylene at these temperatures. Dashed lines: JetSurF 1.0; heavy solid lines: updated JetSurF 1.0; light solid lines: current measurements. The updated JetSurF 1.0 and the JetSurF 1.0 simulations are indistinguishable at the two lower temperatures.

At high temperatures, *n*-alkanes rapidly decompose, via several decomposition pathways, into several stable olefins including ethylene, propene and butenes. Thus, ethylene yields can be used to test predictions of fuel decomposition pathway branching ratios. Here we have defined ethylene yield as the ratio of the peak ethylene and the initial fuel concentrations. Figure 9 presents the ethylene yields during *n*-heptane pyrolysis as a function of temperature. A small increase in the ethylene yield with temperature (from 2.2 to

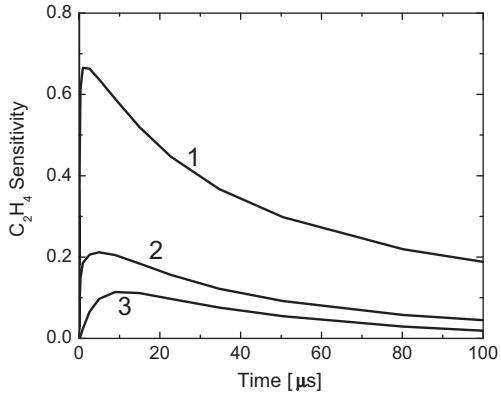

Fig. 8. Ethylene sensitivity analysis during *n*-heptane pyrolysis for conditions of example 2 in Fig. 7. Analysis using JetSurF 1.0 mechanism. The three reactions with the largest sensitivities are shown: Reaction 1: $n\text{-}C_7H_{16} = C_4H_7 + C_3H_7$; Reaction 2: $n\text{-}C_7H_{16} = C_5H_{11} + C_2H_5$; Reaction 3: $n\text{-}C_7H_{16} + H = C_7H_{15} + H_2$.

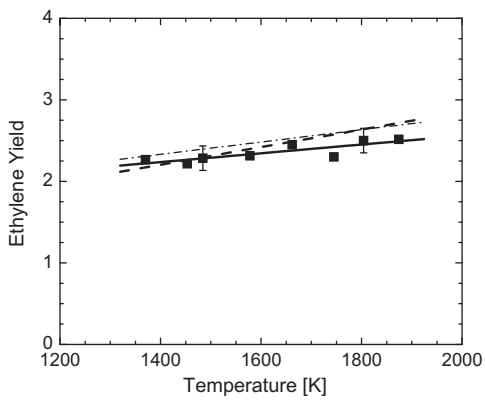

Fig. 9. Ethylene formation yields during *n*-heptane pyrolysis. Yields based on experiments with initial mixtures of 300 ppm *n*-heptane in argon at 1.5–3.7 atm. Solid squares and line: current study; dashed line: updated JetSurF 1.0 simulations; dot–dashed line: JetSurF 1.0.

2.5) is evident as the temperature increases from 1350 to 1850 K (with an uncertainty of ±0.15). Similar ethylene yields were obtained with both the JetSurF 1.0 and the updated JetSurF 1.0 mechanisms.

5. Conclusions and future work

Measurements of ethylene time-histories during ethylene and *n*-heptane pyrolysis were performed using 10.532 μm CO_2 laser absorption spectroscopy. The ethylene absorption cross-section was characterized between 300 and 1950 K using FTIR measurements at low temperature and shock tube measurements at higher temperatures. This technique was then applied to ethylene pyrolysis measurements at temperatures between 1392 and 1981 K. Comparison with Chemkin constant *U–V* simulations using the Marinov et al. kinetics mechanism leads to a revised rate constant (k_1) for the decomposition reaction $C_2H_4 = C_2H_2 + H_2$ near 3 atm. Use of the updated rate constant k_1 in both the Marinov et al. and the Sirjean et al./JetSurF 1.0 mechanisms leads to an improvement in the prediction of the ethylene removal rates. Ethylene time-histories were also measured during *n*-heptane pyrolysis between 1247 and 1874 K. Comparison of these time-history measurements with the predictions of an updated JetSurF 1.0 mechanism showed good agreement with measured ethylene yields, and improved agreement with ethylene removal rates.

Later time ethylene pyrolysis time-histories show discrepancies with current models. Further experiments and analysis are needed to explain these differences. The relatively good agreement between experiment and modeling in the case of *n*-heptane pyrolysis is encouraging evidence of the correctness of the *n*-heptane decomposition pathways and branching ratios in these mechanisms. Further experiments are planned to apply this method to other pyrolysis and oxidation systems including *n*-dodecane and methylcyclohexane, two important jet-fuel surrogate components, as well as diesel fuel, and to expand these time history measurements to include higher alkenes (i.e. propene and butene).

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References

[1] H.J. Curran, P. Gaffuri, W.J. Pitz, C.K. Westbrook, *Combust. Flame* 114 (1998) 149–177.
[2] C.K. Westbrook, W.J. Pitz, O. Herbinet, H.J. Curran, E.J. Silke, *Combust. Flame* 156 (2009) 181–199.
[3] H.J. Curran, P. Gaffuri, W.J. Pitz, C. K Westbrook, *Combust. Flame* 129 (2002) 253–280.
[4] H. Wang, M. Frenklach, *Combust. Flame* 110 (1997) 173–221.
[5] A. Goldaniga, T. Faravelli, E. Ranzi, *Combust. Flame* 122 (2000) 350–358.
[6] P. Berta, S.K. Aggarwal, I.K. Puri, *Combust. Flame* 145 (2006) 740–764.
[7] A.E. Klingbeil, J.B. Jeffries, D.F. Davidson, R.K. Hanson, *Appl. Phys. B* 93 (2008) 627–638.
[8] L. Giroux, M.H. Back, R.A. Back, *Appl. Phys.* B49 (1989) 307–313.

[9] S.W. Sharpe, T.J. Johnson, R.L. Sams, P.M. Chu, G.C. Rhoderick, P.A. Johnson, *Appl. Spectrosc.* 58 (2004) 452–1461.
[10] C.H. Lambeau, A. Fayt, J.L. Duncan, T. Nakagawa, *J. Mol. Spectrosc.* 81 (1980) 227–247.
[11] R.R. Patty, G.M. Russwurm, W.A. McClenny, D.R. Morgan, *Appl. Opt.* 13 (1974) 2850–2854.
[12] R.J. Brewer, C.W. Bruce, J.L. Mater, *Appl. Opt.* 21 (1982) 4092–4100.
[13] A. Mayer, J. Comera, H. Charpentier, C. Jaussaud, *Appl. Opt.* 17 (1978) 391–393.
[14] U. Persson, B. Marthinsson, J. Johansson, S.T. Eng, *Appl. Opt.* 19 (1980) 1711–1715.
[15] J.C. Peterson, M.E. Thomas, R.J. Nordstrom, E.K. Damon, R.K. Long, *Appl. Opt.* 18 (1979) 834–841.
[16] R.J. Kee, F.M. Rupley, J.A. Miller, The Chemkin Thermodynamic Data Base, Report No. SAND-87-8215B, Sandia National Labs., Livermore, CA, USA.
[17] W.J. Witteman, *The CO_2 Laser, Springer Series in Optical Sciences*, vol. 53, Springer-Verlag, New-York, 1987.
[18] N.M. Marinov, W.J. Pitz, C.K. Westbrook, A.M. Vincitore, M.J. Castaldi, S.M. Senkan, *Combust. Flame* 114 (1998) 192–213.
[19] L.S. Zelson, D.F. Davidson, R.K. Hanson, *J. Quant. Spectrosc. Radiat. Transfer* 52 (1993) 31–43.
[20] Th. Just, P. Roth, R. Damm, *Proc. Combust. Inst.* 16 (1977) 961–969.
[21] J.H. Kiefer, S.A. Dapsalis, M.Z. Al-Alami, K.A. Budach, *Combust. Flame* 51 (1983) 79–93.
[22] D.C. Horning, D.F. Davidson, R.K. Hanson, *J. Prop. Power* 18 (2002) 363–371.
[23] B. Sirjean, E. Dames, D.A. Sheen, et al., available at <http://melchior.usc.edu/JetSurF/Version1_0/Index.html>.